

Effect of Pharmacist Intervention and Initiation of Home Blood Pressure Monitoring in Patients with Uncontrolled Hypertension

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This prospective, randomized, controlled study evaluated the impact of pharmacist-initiated home blood pressure monitoring and intervention on blood pressure control, therapy compliance, and quality of life (QOL). Subjects were 36 patients with uncontrolled stage 1 or 2 hypertension. Eighteen subjects received home blood pressure monitors, a diary, and instructions to measure blood pressure twice every morning. Home measurements were evaluated by a clinical pharmacist by telephone, and the patient's family physician was contacted with recommendations if mean monthly values were 140/90 mm Hg or higher. Eighteen control patients did not receive home monitors or pharmacist intervention. Office blood pressure measurements and QOL surveys (SF-36) were obtained at baseline and after 6 months. Mean absolute reductions in systolic and diastolic pressures were significantly reduced from baseline in intervention subjects (17.0 and 10.5 mm Hg, both $p < 0.0001$) but not in controls (7.0 and 3.8 mm Hg, $p = 0.12$ and $p = 0.09$). More intervention subjects (8) had blood pressure values below 140/90 at 6 months compared with controls (4). During the study 83.3% (15) of intervention subjects had drug therapy changes versus 33% (6) of controls ($p < 0.01$). Compliance and QOL were not significantly affected. Our data suggest that the combination of pharmacist intervention with home monitoring can improve blood pressure control in patients with uncontrolled hypertension. This may be related to increased modifications of drug regimens.

(Pharmacotherapy 2000;20(11):1384–1389)

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Supported by the 1998–1999 Bristol-Myers Squibb Pharmacy Practice Hypertension Program grant from the American Association of Colleges of Pharmacy.

Presented at the annual meeting of the American Association of Colleges of Pharmacy, Boston, Massachusetts, July 6, 1999; and the annual meeting of the American College of Clinical Pharmacy, Kansas City, Missouri, October 25, 1999.

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Most patients with hypertension are prescribed drugs to control the disease. Despite such therapy, however, most have inadequate control, with increased risk of associated cardiovascular morbidity and mortality.^{1–4} Moreover, reports from the Minnesota Heart Study suggest that control of hypertension declines with age.^{5–7} The Healthy People 2010 program aims to reach the goal of hypertension control in 50% of hypertensives by the year 2010.⁸ Changes are necessary to improve the care of these patients.¹

Monitoring blood pressure outside the clinic setting commonly is incorporated into comprehensive management of patients with hypertension.

Theoretically, self-measured readings may be more accurate in describing overall blood pressure control than office-based values. Benefits of home monitoring, according to a panel of experts convened by the American Society of Hypertension, include differentiation of sustained hypertension from elevations due to apprehension in the physician's office, assessment of response to antihypertensive drugs, improved compliance with therapy, and possible reduced treatment costs.⁹

Cross-sectional studies suggest that blood pressure values obtained with home monitoring devices may correlate more closely with end-organ damage than office measurements.^{10, 11} A population-based study in Japan also indicated that home measurements are more predictive of mortality than occasional office measurements and show a linear association between home systolic pressure and mortality.¹²

Home monitoring allows patients to participate actively in managing the disease. Although investigators suggested that it may improve compliance, these findings are not consistent.^{13–16} Moreover, the impact of monitoring on disease control and quality of life (QOL) are inconclusive.^{13, 17–20}

Studies have shown that pharmacist involvement in the care of hypertensive patients improves blood pressure control.^{9, 21, 22} These practice models, however, relied on office (or pharmacy) values and did not include home monitoring. The purpose of this study was to evaluate the impact of pharmacist-initiated and -supervised home monitoring on blood pressure control (based on clinic-measured values), compliance with drug therapy, and QOL in patients with uncontrolled hypertension.

Methods

This was a prospective, randomized, controlled study. Patients were recruited from a family medicine residency training clinic in which a pharmacist faculty member provides direct patient care clinical pharmacy services. The University of Colorado institutional review board approved the study protocol and consent form.

Subjects

Patients were reviewed for study eligibility if they were being treated for hypertension and two mercury manometer clinic blood pressures showed stage 1 or 2 hypertension (systolic pressure 140–179 mm Hg and/or diastolic pressure 90–109 mm Hg), as defined by the sixth

report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC-VI).⁴ Inclusion criteria were age 35 years or older, current therapy with at least one antihypertensive drug, stage 1 or 2 hypertension, ability to measure blood pressure with a home monitor, and provision of written informed consent. Exclusion criteria were stage 3 hypertension (systolic ≥ 180 mm Hg and/or diastolic pressure ≥ 110 mm Hg based on JNC-VI criteria), an identified secondary cause of hypertension, atrial fibrillation, pregnancy, current home blood pressure monitoring, failure to demonstrate correct use of monitoring device, and drug or alcohol abuse.

Design

Subjects were randomized using a deck of cards and enrolled in either the intervention or control group. All subjects received counseling on antihypertensive drug therapy and lifestyle modification by one of three investigators using an identical format according to study protocol. During a 30-minute appointment, the pharmacist provided all subjects with written pamphlets from the American Heart Association that explained hypertension and cardiovascular risks. Subjects were counseled on the importance of exercise and salt and alcohol reduction. At this first appointment, as well as the final 6-month appointment, blood pressure was measured (two values separated by at least 30 seconds) using the same mercury sphygmomanometer (Tycos, Arden, NC). At both visits each subject completed a Medical Outcomes Survey-Short Form 36 (SF-36) on QOL.²³

Intervention subjects were given a regular-size AnD (Milpitas, CA) UA-702 manual electronic blood pressure monitor and instructed on its proper use. If arm circumference was greater than 12 inches, a large adult-size cuff was given. This model was chosen because it was the highest-rated manual model based on accuracy, consistency, and ease of use and instructions according to Consumer Reports in 1996.²⁴ Subjects were instructed to monitor blood pressure each morning, before food, coffee, or drugs, after a 5-minute rest in a seated position, and again after 2–5 minutes. Each subject was given a predated diary in which they documented the two morning values, changes in antihypertensive drug therapy, and missed doses.

A clinical pharmacist contacted each intervention

Table 1. Baseline Demographics

Characteristic	Control Group (n=18)	Intervention Group (n=18)
Age (yrs)	57.6 ± 13.5	60.0 ± 14.8
M/F	7/11	4/14
Systolic blood pressure (mm Hg)	153.9 ± 14.6	157.9 ± 16.4
Diastolic blood pressure (mm Hg)	89.6 ± 9.8	91.1 ± 10.8
Mean arterial pressure (mm Hg)	111.0 ± 6.4	113.4 ± 8.0
Race		
Caucasian	15	13
African-American	2	3
Hispanic	1	1
Asian	0	1
Target organ disease (no.)	1	2
Cardiac risk factors ^a		
Smoker	5	2
Dyslipidemia	3	7
Diabetes mellitus	4	3
Age > 60 yrs	5	10
Family history	3	5

^aAs defined by JNC-VI.⁴

p<0.05 for all values.

subject by telephone after 1 month to evaluate blood pressure response. If mean monthly home values were above 140/90 mm Hg (as calculated by an investigator), primary care physicians were informed and therapy recommendations were made as necessary. These subjects were contacted again by telephone at monthly intervals to evaluate average blood pressure measurements and response to therapy. If mean monthly blood pressure remained above 140/90, they were contacted at 2-month intervals for follow-up evaluations.

There were no restrictions on how often patients had office visits with primary care providers during the study. Frequency of office visits was at the discretion of the physicians.

Control subjects were not given a home blood pressure monitor. They were allowed to continue monitoring at retail stores (pharmacy, grocery store) or at the medical clinic if they already were doing so. However, to maintain control conditions, they agreed not to start home monitoring during the 6 months of study. Primary care providers continued to make antihypertensive drug adjustments as part of routine care. Similar to the intervention group, controls were not restricted as to frequency of office visits with primary care providers.

Six months after randomization and enrollment, intervention and control patients returned to the clinic and two blood pressures were measured with a mercury sphygmomanometer. Intervention subjects returned their

diaries at this time. Control patients were given a home blood pressure monitor and intervention subjects were allowed to keep their monitors as compensation. All subjects completed a second SF-36 survey. Pharmacies were contacted at the end of the study to obtain antihypertensive prescription refill data. Percentage compliance was calculated by dividing the number of tablets/capsules refilled by the amount prescribed during the study.

Statistical Analysis

For outcome measures of continuous data, paired *t* tests were used for within-group comparisons, and unpaired *t* tests for between-group comparisons. The χ^2 test was used to analyze dichotomous data. The SAS statistical package (SAS, Cary, NC) with standard methodology was used for SF-36 data analysis. Differences were considered statistically significant at a *p* value less than 0.05. Data are presented as mean ± standard deviation (SD) unless stated otherwise.

Results

Subject Characteristics

Approximately 80 family medicine patients with clinic systolic pressure above 140 mm Hg and/or diastolic pressure above 90 mm Hg were evaluated over a 12-week period. Forty-one met study inclusion criteria and were enrolled. Five

subjects (3 controls, 2 intervention) were excluded from the final analysis: one control was lost to follow-up, one moved to assisted living 2 months after enrollment, and one had atrial fibrillation; one intervention patient moved out of the country, and one did not receive drug therapy at the time of enrollment. Statistical analysis revealed no statistically significant differences in baseline characteristics between control and intervention groups (Table 1). Several subjects had known risk factors for cardiovascular disease, and three had established target organ disease (1 heart failure, 1 coronary artery disease, 1 hypertension-associated retinopathy).

Blood Pressure Response

Within-group analysis revealed that intervention patients had significant mean reductions from baseline in systolic and diastolic pressures and mean arterial pressure (MAP) of 17.1, 10.5, and 12.7 mm Hg respectively (all $p < 0.001$). The only significant reduction from baseline in control subjects was 4.9 mm Hg in MAP ($p = 0.049$). Between-group analysis showed significant differences in reductions in follow-up clinic diastolic pressure and MAP from baseline ($p = 0.022$ and 0.010 , respectively). Despite a trend that suggested greater reduction in mean systolic pressure in the intervention group, this difference failed to reach statistical significance ($p = 0.069$; Figure 1). At 6-month follow-up, only 22% (4/18) of controls had follow-up clinic blood pressure measurements (average of two measurements) below 140/90, compared with 44% (8/18) of intervention patients ($p > 0.1$).

Twelve diaries from intervention subjects were collected at 6-month follow-up; the other 6 were claimed to be lost. Two subjects who submitted a

diary could not complete 6 months of home monitoring but were included in the final analysis. One experienced intolerable anxiety associated with daily blood pressure measurements and discontinued after 10 weeks; the other could not continue measurements because of a hand tremor.

Subjects measured blood pressure at home for a median of 184 days (range 121–196 days). They did so twice in the morning during 87.6% of these days. At least one measurement was collected during 89.9% of monitoring days. First morning values were significantly higher than second values ($p < 0.001$ systolic and diastolic). Mean differences between first and second values were 4.1 ± 9.16 mm Hg for systolic and 1.8 ± 6.57 mm Hg for diastolic pressures.

Secondary Outcomes

Mean compliance with antihypertensive drug therapy was 89% in the control group and 82% in the intervention group ($p = 0.29$). No intervention subjects documented missed doses in the diaries. Throughout the study, 83% (15/18) of intervention patients had changes in antihypertensive drugs. These consisted of a dosage increase, addition of another agent, or switching to an alternative agent. Only 33% (6/18) of controls had antihypertensive drugs changed ($p < 0.01$; Table 2). Intervention subjects had fewer mean office visits with primary care providers during the 6 months than controls, but this was not statistically significant (2.72 vs 4.44 mean office visits/patient, $p = 0.08$).

The SF-36 results found no statistical difference between groups at baseline and at 6-month follow-up (Table 3) except for bodily pain at baseline (lower in the control group, $p = 0.04$). No significant changes from baseline values were observed in within-group analyses (all comparisons, $p > 0.1$).

Discussion

Patients with uncontrolled hypertension who measured their blood pressure with a home monitoring device in combination with pharmacist intervention had significant improvements in blood pressure control. Our results are consistent with other studies that reported the positive influence of pharmacists on patient blood pressure control when they are members of the health care team.^{21, 22} Antihypertensive drugs were adjusted more frequently in the intervention group (83% vs 33%

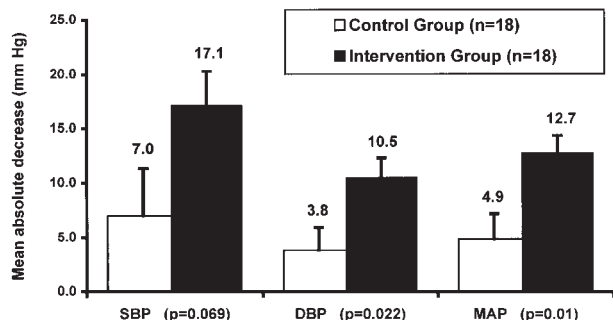


Figure 1. Between-group analysis of mean absolute decreases (mm Hg) in systolic and diastolic pressures and MAP from baseline based on clinic mercury sphygmomanometer readings. Error bars = SEM.

Table 2. Antihypertensive Therapy at Baseline and after 6 Months

Antihypertensive Agents	Control Group		Intervention Group	
	Baseline	6 Months	Baseline	6 Months
α -Blockers	1	2	0	0
ACE inhibitors	8	8	9	8
Angiotensin II receptor blockers	0	0	0	1
β -Blockers	10	9	6	8
Calcium channel blockers	4	3	6	5
Diuretics	11	12	12	13
Others	1	1	0	0

ACE = angiotensin-converting enzyme.

Data are presented as number of patients receiving a drug from each class.

Table 3. MOS-SF 36 Health-Related Quality of Life Scores (range 0–100)

Domain	Baseline		6 Months	
	Control Group	Intervention Group	Control Group	Intervention Group
Physical function	65.8	66.6	71.1	67.4
Role physical	48.6	68.1	58.3	57.8
Bodily pain ^a	46.1	61.1	54.0	62.5
General health	60.1	67.9	57.4	62.8
Vitality	54.4	50.6	50.6	53.2
Social function	70.1	82.6	71.5	78.7
Role-emotional	70.4	75.9	62.7	72.5
Mental health	71.3	71.3	66.9	75.3

^aDifference between groups at baseline was $p=0.04$.

$p>0.1$ for all within-group comparisons.

controls) despite the trend that control subjects visited primary care physicians more frequently. We believe this was a direct result of pharmacist interventions that alerted physicians to uncontrolled blood pressure values, providing recommendations and facilitating therapy adjustments. Our results suggest that all of these components can improve blood pressure control.

To our knowledge, no published studies have examined the impact of pharmacist intervention in combination with home blood pressure monitoring on blood pressure control. However, several studies reported improvements in blood pressure control with an interdisciplinary team approach that involved physicians, pharmacists, and nurses.^{21, 22} Pharmacists are members of structured health care teams in academic health centers, Veterans Affairs medical centers, and some managed care organizations. In addition, some community pharmacists work collaboratively with physicians to manage chronic diseases including hypertension.²⁵ Teaching patients how to use home blood pressure monitors as well as assessing and following patients' progress offers pharmacists the opportunity to have an impact on hypertension management and may improve patient care.

An interesting finding in this study was a

significant difference between first and second blood pressure measurements. Although JNC-VI guidelines recommend taking and averaging two or more readings (separated by at least 2 min), this is not always followed in practice. Our results reinforce the importance of evaluating at least two consecutive measurements at home or clinic when determining the need to adjust antihypertensive drugs, as the second is often lower.

Patient compliance and QOL were not affected in our study. Compliance rates in both groups were similar and high (> 80%). This may be explained in part because both groups received initial counseling by the clinical pharmacist after enrollment. Several clinical trials evaluated the effect of home monitoring on blood pressure and similarly found that whereas clinic values may be improved, compliance with drug therapy is not.^{15, 25, 26}

Our study had several limitations. Because the population was small, it lacks statistical power to detect a significant difference in baseline and follow-up systolic pressures in intervention patients. In addition, subjects were recruited from one family medicine clinic, so extrapolating our results to other clinic settings and patient populations may be limited. Finally, disease

control was assessed based on the average of two office blood pressure values, which can vary based on factors such as time of day.

Observer error in reporting self-measured values also must be considered when using home blood pressure monitors. Patients reportedly fail to document up to 36% of measured values.²⁷ Similarly, our subjects may have underreported home values. This should be taken into account whenever patient-reported data are used to influence clinical decisions. However, significant clinic blood pressure reductions at the end of our study for intervention patients suggest that even with possible observer error in reporting, these measurements are useful.

Based on these findings, we conclude that home blood pressure monitoring in combination with clinical pharmacist intervention is an effective method to improve blood pressure control in patients with uncontrolled hypertension. Forty-four percent of intervention patients had controlled hypertension after 6 months, compared with only 22% of controls receiving usual care. Achievement of goal blood pressure values in the intervention group was close to the 50% goal set by Healthy People 2010.⁸ Heightened awareness of the disease as well as increased pharmacist-physician interaction may facilitate drug therapy modifications that are necessary to achieve desired blood pressure goals. Additional studies in expanded primary care populations are required to evaluate benefits and costs of home monitoring as an effective method to improve the control of hypertension.

Acknowledgments

The authors thank Julie Paranka, M.D., for assistance with study design, and Daniel Malone, Ph.D., for assistance with statistical analysis.

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